

SQL PROJECT ON PIZZA SALES

Presented by **Ritu Thakur**



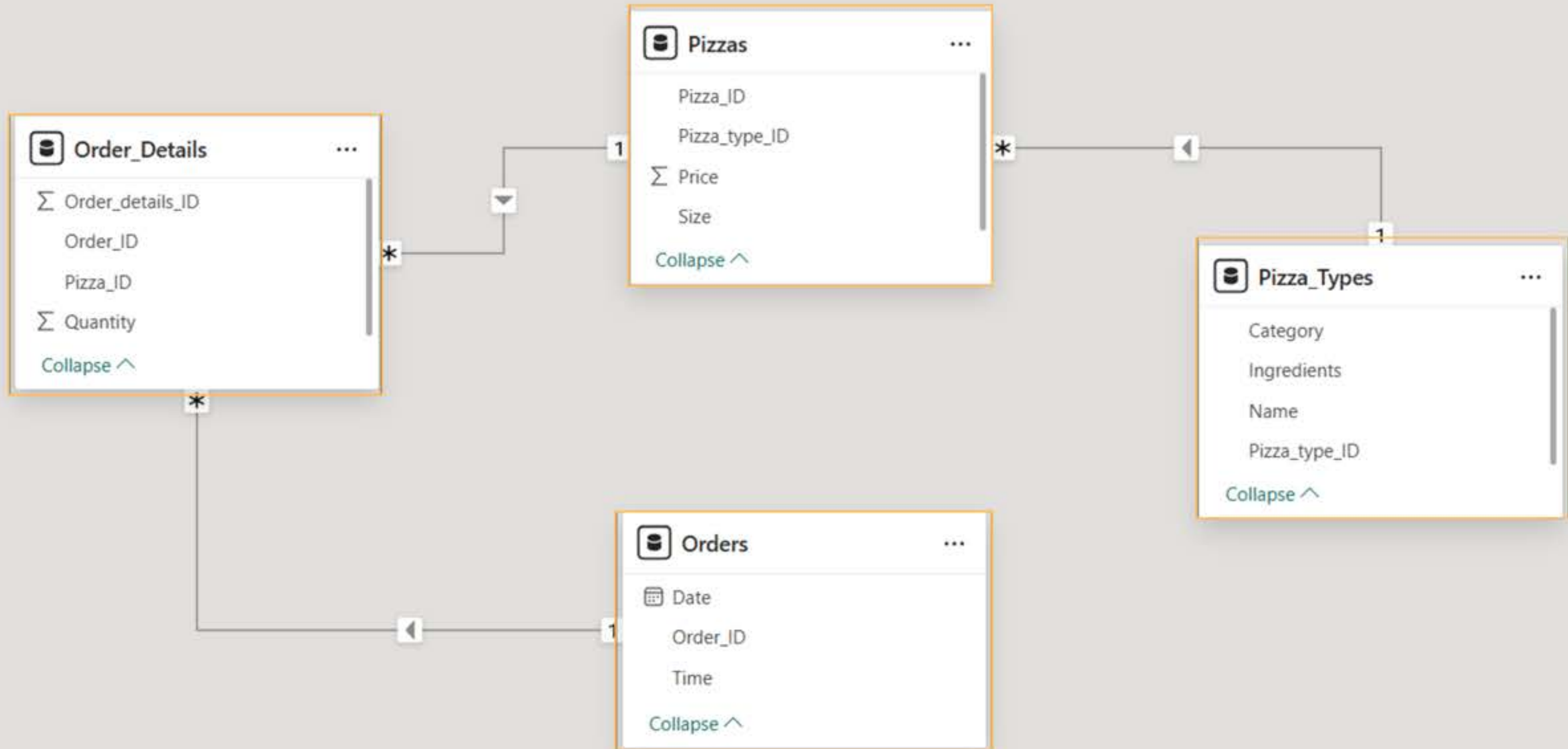
PROJECT OVERVIEW:

Hello, I'm Ritu Thakur, currently developing my skills in data analysis.

In this project, I've explored pizza sales data using SQL to answer business-related questions. Through structured queries, I analyzed customer preferences, sales trends, and revenue performance to gain meaningful insights that can support data-driven decisions.



SCHEMA



QUESTION - SET

BASIC:

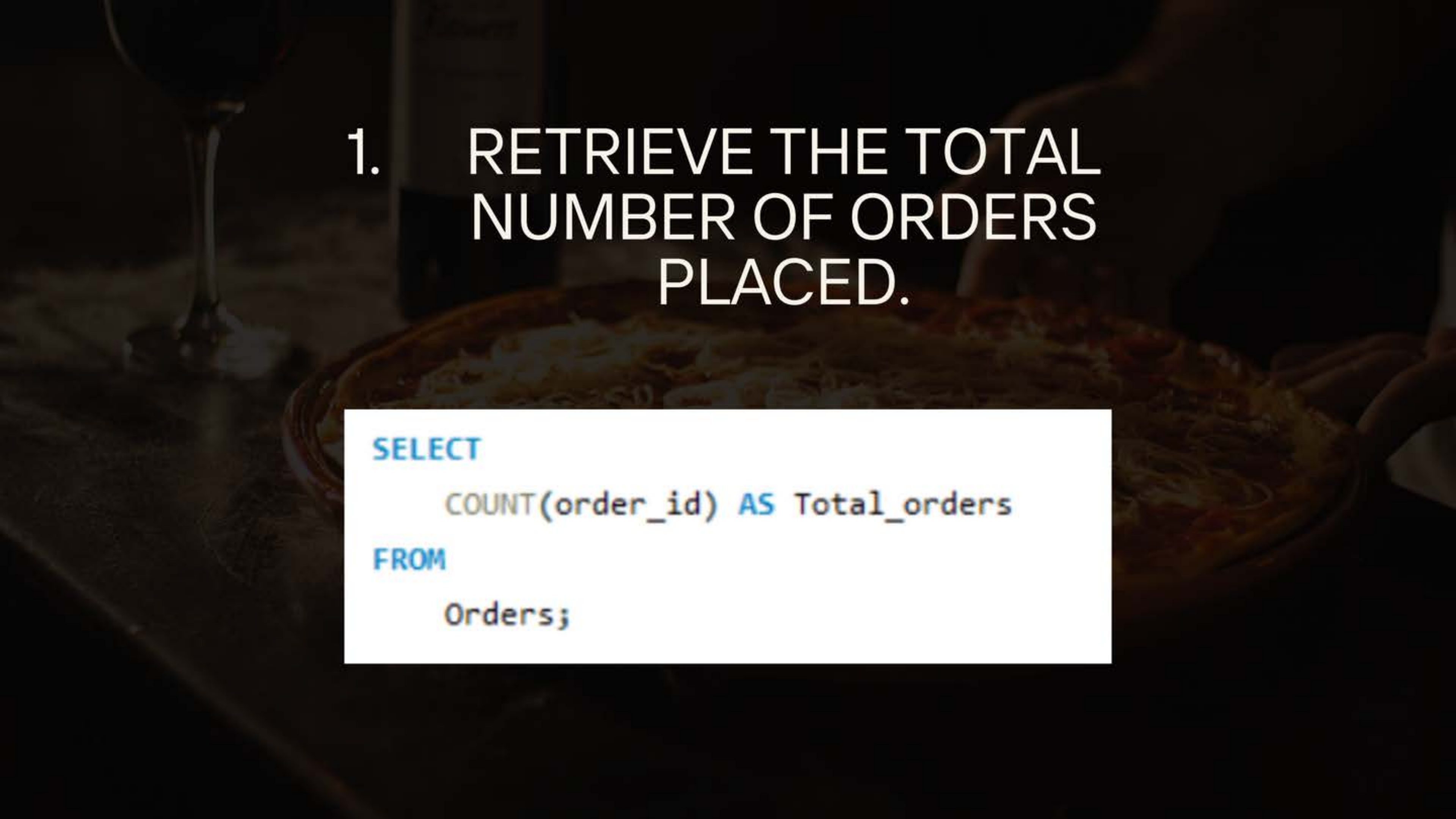
- Retrieve the total number of orders placed.
- Calculate the total revenue generated from pizza sales.
- Identify the highest-priced pizza.
- Identify the most common pizza size ordered.
- List the top 5 most ordered pizza types along with their quantities.

INTERMEDIATE:

- Join the necessary tables to find the total quantity of each pizza category ordered.
- Determine the distribution of orders by hour of the day.
- Join relevant tables to find the category-wise distribution of pizzas.
- Group the orders by date and calculate the average number of pizzas ordered per day.
- Determine the top 3 most ordered pizza types based on revenue.

ADVANCED:

- Calculate the percentage contribution of each pizza type to total revenue.
- Analyze the cumulative revenue generated over time.
- Determine the top 3 most ordered pizza types based on revenue for each pizza category.



1. RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED.

```
SELECT
```

```
    COUNT(order_id) AS Total_orders
```

```
FROM
```

```
    Orders;
```

2. CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES.

```
SELECT
    ROUND(SUM(od.quantity * p.price), 2) AS Total_Sales
FROM
    order_details od
    JOIN
    pizzas p ON od.Pizza_id = p.pizza_id;
```

3. IDENTIFY THE HIGHEST-PRICED PIZZA.

```
SELECT
    pt.name, p.price
FROM
    pizza_types pt
    JOIN
        Pizzas p ON pt.pizza_type_id = p.pizza_type_id
ORDER BY p.price DESC
LIMIT 1;
```


4. IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED.

```
SELECT
    p.size, COUNT(od.order_details_id) AS Order_cnt
FROM
    pizzas p
    JOIN
        order_details od ON p.pizza_id = od.Pizza_id
GROUP BY p.size
ORDER BY order_cnt DESC;
```


5. LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES.

```
SELECT
    pt.name, SUM(od.quantity) AS Quantity
FROM
    pizza_types pt
    JOIN
    pizzas p ON pt.pizza_type_id = p.pizza_type_id
    JOIN
    order_details od ON p.pizza_id = od.Pizza_id
GROUP BY pt.name
ORDER BY Quantity DESC
LIMIT 5;
```

6. JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED.

```
SELECT
    pt.category, SUM(od.quantity) AS Quantity
FROM
    pizza_types pt
    JOIN
    pizzas p ON pt.pizza_type_id = p.pizza_type_id
    JOIN
    order_details od ON p.pizza_id = od.Pizza_id
GROUP BY pt.category
ORDER BY Quantity DESC;
```


7. DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY.

```
SELECT
    HOUR(order_time) AS Hour, COUNT(order_id) AS Order_cnt
FROM
    orders
GROUP BY HOUR(order_time);
```

8. JOIN RELEVANT TABLES TO FIND THE CATEGORY-WISE DISTRIBUTION OF PIZZAS.

```
SELECT
    category, COUNT(name) AS Total_distribution
FROM
    pizza_types
GROUP BY category;
```


9. GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY.

```
SELECT
    ROUND(AVG(Quantity), 0) AS Avg_pizza_ordered_per_day
FROM
    (SELECT
        o.order_date, SUM(od.quantity) AS quantity
    FROM
        orders o
    JOIN order_details od ON o.Order_id = od.Order_id
    GROUP BY o.Order_date) AS order_quantity;
```

10. DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE.

```
SELECT
    Pt.name, SUM(Od.quantity * p.price) AS Revenue
FROM
    pizza_types pt
    JOIN
    pizzas p ON pt.pizza_type_id = p.pizza_type_id
    JOIN
    order_details od ON p.Pizza_id= Od.Pizza_id
GROUP BY Pt.name
ORDER BY Revenue DESC
LIMIT 3;
```


11. CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE.

```
SELECT
    pt.category,
    ROUND(SUM(od.quantity * p.price) / (SELECT
        ROUND(SUM(od.quantity * p.price), 2) AS total_sales
    FROM
        order_details od
    JOIN
        pizzas p ON od.pizza_id = p.pizza_id) * 100,
    2) AS revenue
FROM
    pizza_types pt
    JOIN
    pizzas p ON pt.pizza_type_id = p.pizza_type_id
    JOIN
    order_details od ON od.pizza_id = p.pizza_id
GROUP BY pt.category
ORDER BY revenue DESC;
```

12. ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME.

```
SELECT
    order_date, ROUND(SUM(revenue), 2) AS cum_revenue
FROM
    (SELECT
        o.order_date, SUM(od.quantity * p.price) AS revenue
    FROM
        order_details od
    JOIN pizzas p ON od.pizza_id = p.pizza_id
    JOIN orders o ON o.order_id = od.order_id
    GROUP BY o.order_date) AS sales
GROUP BY order_date;
```


13. DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY.

```
SELECT
    category,
    name,
    revenue
FROM (
    SELECT
        pt.category,
        pt.name,
        SUM(od.quantity * p.price) AS revenue,
        RANK() OVER(PARTITION BY pt.category ORDER BY SUM(od.quantity * p.price) DESC) AS rn
    FROM
        pizza_types pt
    JOIN
        pizzas p ON pt.pizza_type_id = p.pizza_type_id
    JOIN
        order_details od ON p.pizza_id = od.pizza_id
    GROUP BY
        pt.category, pt.name
) AS ranked_pizzas
WHERE rn <= 3;
```


CONCLUSION

This project allowed me to apply core SQL skills to analyze real-world pizza sales data. By exploring customer trends, revenue patterns, and product performance, I gained hands-on experience in data cleaning, querying, and drawing business insights.

Thank you for viewing my project!

Connect with me professionally:

 [LinkedIn – Ritu Thakur](#)

 Explore the full project here:

[GitHub – SQL Projects](#)